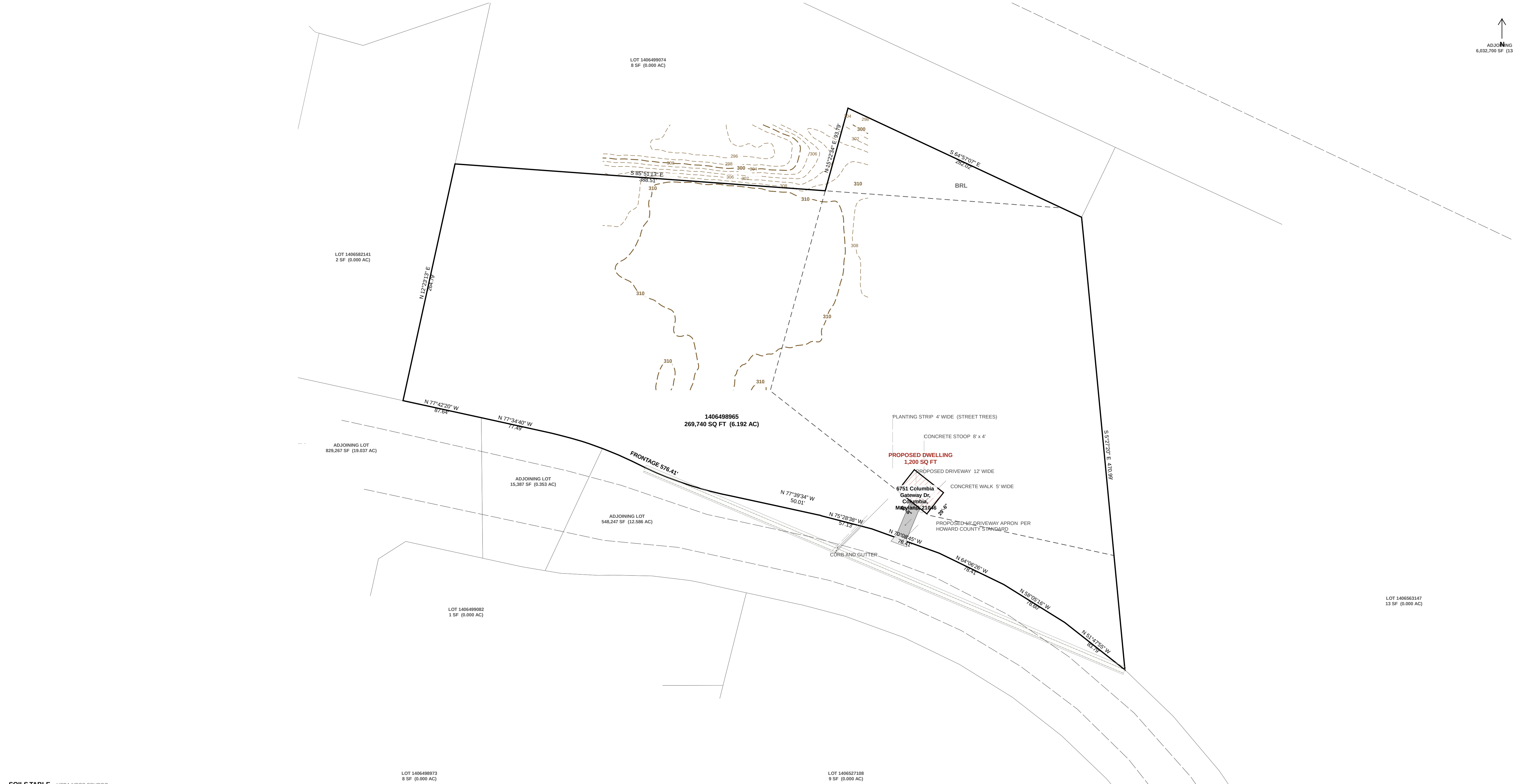


KEALEE

Design. Build. Deliver.

PROJECT
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
ADDRESS
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
JURISDICTION
Howard County, Maryland
ZONE

SHEET
C-100 — Existing Conditions and Boundary Plan
SCALE
1" = 60'

STATUS
PRELIMINARY
SHEET NO.
1 OF 5

COORDINATES
EPSG:2248
H: NAD83
V: NAVD83 (feet)

PROFESSIONAL RESPONSIBILITY
Maryland Licensed Surveyor
certifies: boundary_determination, topographic_survey

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope.

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-100 — Existing Conditions and Boundary Plan
DISCIPLINE	Maryland Licensed Surveyor
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

ZONE		
TAX / PARCEL ID	1406489965	
GROSS TRACT AREA	269,740 SF	6.192 AC
LOT FRONTAGE	—	576.41'
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	—	—
SIDE YARD	—	—
REAR YARD	—	—
COVERAGE	MAX	PROPOSED
LOT COVERAGE	—	0.4%
BUILDING FOOTPRINT	—	1,200 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD83 (feet)	

SITE ANALYSIS

1. Gross tract area	269,740 SF
2. Area of dwelling	1,200 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	269,740 SF
6. TOTAL AREA DISTURBED	13,257 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH HOWARD COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
6751 Columbia Gateway Dr, Columbia, Maryland, 21046	—	—	—	—	20'	2

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total — 0.7% Impervious	269,740	6.192	
PRE-DEVELOPMENT	SF		C
Pervious — lawn / meadow	269,740		0.25
Composite C			0.250
POST-DEVELOPMENT	SF		C
Proposed roof	1,200		0.95
Driveway and on-lot paving	574		0.95
Pervious — lawn	267,966		0.25
Composite C			0.255
Flow path			1001 ft
Flow path slope			1.4 %
Time of concentration (Kirpich)			8.2 min
Rainfall intensity, 10-yr			5.71 in/hr
Peak discharge PHE, Q = CIA			9.84 cfs
Peak discharge POST, Q = CIA			9.02 cfs
INCREASE TO BE MANAGED			0.18 cfs
Rv = 0.05 + 0.0091			0.059
Water quality volume WQv			1,257 cf

Time of concentration by Kirpich over a 1001 ft flow path at 1.4%, taken from the 2 ft contours across the catchment. A 5-minute flow applies.
Rainfall intensity 5.71 in/hr for the 10-year storm at a 6.2 minute duration, NOAA Atlas 14 Vol. 2 Ver. 3, PFDS point estimate, 39.1842 N 76.8016 W.
Water quality volume on a 1.0 in rainfall, Maryland Stormwater Design Manual, Vol. 1 & 2 — WQV on a 1.0 in rainfall.
Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed areas.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION. FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY TEST PIT BEFORE CONSTRUCTION.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS COMPILED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION, NOT AN OPEN DITCH SECTION. GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSED CURB AT THE APRON PER HOWARD COUNTY STANDARD. ONLY LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE

DATE

SEAL

LICENSE NO.

EXPIRATION DATE

REVISIONS

— NONE —

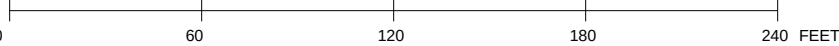
SOILS TABLE

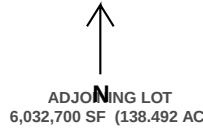
USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP	DRAINAGE CLASS
AwB	Alloway silt loam, 2 to 5 percent slopes	.32	No	D	Moderately well drained
BaA	Balle silt loam, 0 to 3 percent slopes	.37	Yes	C/D	Poorly drained
BeA	Benevola silt loam, 0 to 3 percent slopes	.43	No	C	Well drained
BeB	Benevola silt loam, 3 to 8 percent slopes	.43	No	C	Well drained
BeC	Benevola silt loam, 8 to 15 percent slopes	.43	No	C	Well drained
B/C	Brinklow channery loam, 8 to 15 percent slopes	.28	No	C	Well drained
B/D	Brinklow channery loam, 15 to 25 percent slopes	.28	No	C	Well drained
B/F	Brinklow-Blocktown channery loams, 25 to 65 percent slopes	.32	No	C	Well drained

SOURCE AND ACCURACY
Parcel boundary — Jurisdiction gis — Maryland Department of Planning — MD MAP parcel boundaries — Plat...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 60'





SHEET
C-200 — Overall Site and Zoning Plan
SCALE
1" = 60'
STATUS
PRELIMINARY
SHEET NO.
2 OF 5
COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

SEAL AND SIGNATURE

Architect
certifies: architectural_footprint

SEAL AND SIGNATURE

This sheet carries content from more than one professional discipline. Each seal below applies only to the subjects listed beside it. No professional certifies content outside their stated scope, and the sheet is not divided into separate drawings for that purpose.

[illegible]

SITE ANALYSIS	
1. Gross tract area	269,740 SF
2. Area of dwelling	1,200 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	269,740 SF
6. TOTAL AREA DISTURBED	13,257 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH HOWARD COUNTY

DRAINAGE AREA COMPUTATIONS			
CATCHMENT	SF	ACRES	C
Total = 0.74 acres	269,740	6.192	
PRE-DEVELOPMENT	SF		
Pervious = lawn / meadow	269,740		0.25
Composite C			0.250
POST-DEVELOPMENT	SF		
Proposed roof	1,200		0.95
Driveway and on lot paving	574		0.95
Pervious = lawn	267,966		0.25
Composite C			0.355
Flow path			1001 ft
Flow path slope			1.4%
Time of concentration (Kirpich)			8.2 min
Rainfall intensity, 10-yr			5.71 in/hr
Peak discharge PRE, Q = CIA			8.84 cfs
Peak discharge POST, Q = CIA			9.02 cfs
INCREASE TO BE ADDED			0.18 cfs
Rv = 0.05 - 0.091			0.0559
Water quality volume WQv			1,257 cf

SEQUENCE OF CONSTRUCTION

- 1 DAY
2 DAYS
1 DAY
2 DAYS
5 DAYS
MONTHS
10 DAYS
5 DAYS
1 DAY

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

PROFESSIONAL CERTIFICATION

SEAL	SIGNATURE _____	DATE _____
	LICENSE NO. _____	EXPIRATION DATE _____

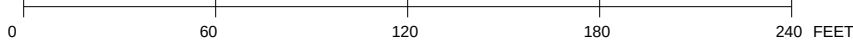
REVISIONS

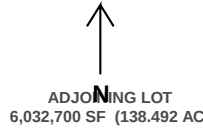
— none —

SOILS TABLE USDA NRCS SSURGO

SOURCE AND ACCURACY
Parcel boundary — jurisdiction gis — Maryland Department of Planning — MD iMAP parcel boundaries — Plat..
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 60'





KEALEE

Design, build, deliver

PROJECT
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
ADDRESS
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
JURISDICTION
Howard County, Maryland
ZONE

SHEET
C-400 — Grading and Drainage Plan
SCALE
1" = 60'

STATUS
PRELIMINARY
SHEET NO.
3 OF 5

COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

PROFESSIONAL RESPONSIBILITY
Maryland Professional Engineer
certifies: grading_design, sediment_control, stormwater_design

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope.

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-400 — Grading and Drainage Plan
DISCIPLINE	Maryland Professional Engineer
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

ZONE		
TAX / PARCEL ID	1406498965	
GROSS TRACT AREA	269,740 SF	6.192 AC
LOT FRONTAGE	—	576.41'
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	—	—
SIDE YARD	—	—
REAR YARD	—	—
COVERAGE	MAX	PROPOSED
LOT COVERAGE	—	0.4%
BUILDING FOOTPRINT	—	1,200 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	269,740 SF
2. Area of dwelling	1,200 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	269,740 SF
6. TOTAL AREA DISTURBED	13,257 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH HOWARD COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
6751 Columbia Gateway Dr, Columbia, Maryland, 21046	—	—	—	—	20'	2

6751 Columbia Gateway Dr, Columbia, Maryland, 21046
G: ground level, B: basement, FF: finished floor, SF: subfloor, HT: height, STY: storeys. A DASH IS NOT ZERO: the elevation has not been established and must be set from a field-run survey or construction.

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total — 0.7% Impervious	269,740	6.192	—
PRE-DEVELOPMENT	SF	—	C
Pervious — lawn / meadow	269,740	—	0.25
Composite C	—	—	0.250
POST-DEVELOPMENT	SF	—	C
Proposed roof	1,200	—	0.95
Driveway and on-lot paving	574	—	0.95
Pervious — lawn	267,966	—	0.25
Composite C	—	—	0.255
Flow path	1001 ft	—	—
Flow path slope	1.4 %	—	—
Time of concentration (Kirpich)	8.2 min	—	—
Rainfall intensity, 10-yr	5.71 in/hr	—	—
Peak discharge PRE, Q = CIA	9.84 cfs	—	—
Peak discharge POST, Q = CIA	9.02 cfs	—	—
INCREASE TO BE MANAGED	0.18 cfs	—	—
Rv = 0.05 + 0.0091	0.059	—	—
Water quality volume WQv	1,257 cf	—	—

Time of concentration by Kirpich over a 1001 ft flow path at 1.4%, taken from the 2 ft contours across the catchment. A 5-minute flow applies.
Rainfall intensity 5.71 in/hr for the 10-year storm at a 6.2 minute duration, NOAA Atlas 14 Vol. 2 Ver. 3, PFDS point estimate, 39.1842 N 76.8016 W.
Water quality volume on a 1.0 in rainfall, Maryland Stormwater Design Manual, Vol. 1 & 2 — WQv on a 1.0 in rainfall.
Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed areas.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION. FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY TEST PIT BEFORE CONSTRUCTION.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS COMPILED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED. THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION, NOT AN OPEN DITCH SECTION. GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSED CURB AT THE APRON PER HOWARD COUNTY STANDARD. ONLY LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE

DATE

SEAL

LICENSE NO.

EXPIRATION DATE

REVISIONS

— NONE —

SOILS TABLE

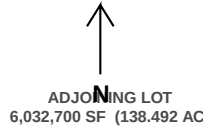
USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP.	DRAINAGE CLASS
AwB	Alloway silt loam, 2 to 5 percent slopes	.32	No	D	Moderately well drained
BaA	Baile silt loam, 0 to 3 percent slopes	.37	Yes	C/D	Poorly drained
BeA	Benevola silt loam, 0 to 3 percent slopes	.43	No	C	Well drained
BeB	Benevola silt loam, 3 to 8 percent slopes	.43	No	C	Well drained
BeC	Benevola silt loam, 8 to 15 percent slopes	.43	No	C	Well drained
B/C	Brinklow channery loam, 8 to 15 percent slopes	.28	No	C	Well drained
B/D	Brinklow channery loam, 15 to 25 percent slopes	.28	No	C	Well drained
B/F	Brinklow-Blocktown channery loams, 25 to 65 percent slopes	.32	No	C	Well drained

SOURCE AND ACCURACY
Parcel boundary — Jurisdiction gis — Maryland Department of Planning — MD MAP parcel boundaries — Plat...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 60'

0 60 120 180 240 FEET



KEALEE

Design, build, deliver

PROJECT
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
ADDRESS
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
JURISDICTION
Howard County, Maryland
ZONE

SHEET
C-500 — Utility Plan
SCALE
1" = 60'
STATUS
PRELIMINARY
SHEET NO.
4 OF 5

COORDINATES
EPSG-2248
H: NAD83
V: NAVD88 (feet)

PROFESSIONAL RESPONSIBILITY
Maryland Professional Engineer
certifies: Utility_Design

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope.

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-500 — Utility Plan
DISCIPLINE	Maryland Professional Engineer
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

ZONE	
TAX / PARCEL ID	1406488965
GROSS TRACT AREA	269,740 SF 6.192 AC
LOT FRONTAGE	— 576.41'
SETBACKS	REQUIRED PROVIDED
FRONT YARD	— —
SIDE YARD	— —
REAR YARD	— —
COVERAGE	MAX PROPOSED
LOT COVERAGE	— 0.4%
BUILDING FOOTPRINT	— 1,200 SF
DATUM	
HORIZONTAL	EPSG-2248 NAD83
VERTICAL	NAVD88 (feet)

SITE ANALYSIS

1. Gross tract area	269,740 SF
2. Area of dwelling	1,200 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	269,740 SF
6. TOTAL AREA DISTURBED	13,257 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH HOWARD COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
6751 Columbia Gateway Dr, Columbia, Maryland, 21046	—	—	—	—	20'	2

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total — 0.7% Impervious	269,740	6.192	
PRE-DEVELOPMENT	SF		C
Pervious — lawn / meadow	269,740	0.25	
Composite C		0.250	
POST-DEVELOPMENT	SF		C
Proposed roof	1,200	0.05	
Driveway and on-lot paving	574	0.05	
Pervious — lawn	267,866	0.25	
Composite C		0.255	
Flow path		1001 ft	
Flow path slope	1.4 %		
Time of concentration (Kirpich)	8.2 min		
Rainfall intensity, 10-yr	5.71 in/hr		
Peak discharge PRE, Q = CIA	9.84 cfs		
Peak discharge POST, Q = CIA	9.02 cfs		
INCREASE TO BE MANAGED	0.18 cfs		
Rv = 0.05 + 0.0091	0.059		
Water quality volume WQv	1,257 cf		

Time of concentration by Kirpich over a 1001 ft flow path at 1.4%, taken from the 2 ft contours across the catchment. A 5-minute flow applies.
Rainfall intensity 5.71 in/hr for the 10-year storm at a 6.2 minute duration. NOAA Atlas 14 Vol. 2 Ver. 3. PFDS point estimate, 39.1842 N 76.8016 W.
Water quality volume on a 1.0 in rainfall. Maryland Stormwater Design Manual, Vol. 1 & 2 — WQv on a 1.0 in rainfall.
Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed areas.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION. FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY TEST PIT BEFORE CONSTRUCTION.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS COMPILED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION. NOT AN OPEN DITCH SECTION. GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSED CURB AT THE APRON PER HOWARD COUNTY STANDARD. ONLY LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE

DATE

SEAL

LICENSE NO.

EXPIRATION DATE

REVISIONS

— NONE —

SOILS TABLE

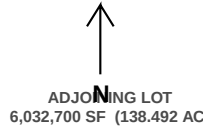
USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP.	DRAINAGE CLASS
AwB	Alloway silt loam, 2 to 5 percent slopes	.32	No	D	Moderately well drained
BaA	Baile silt loam, 0 to 3 percent slopes	.37	Yes	C/D	Poorly drained
BeA	Benevola silt loam, 0 to 3 percent slopes	.43	No	C	Well drained
BeB	Benevola silt loam, 3 to 8 percent slopes	.43	No	C	Well drained
BeC	Benevola silt loam, 8 to 15 percent slopes	.43	No	C	Well drained
B/C	Brinklow channery loam, 8 to 15 percent slopes	.28	No	C	Well drained
B/D	Brinklow channery loam, 15 to 25 percent slopes	.28	No	C	Well drained
B/F	Brinklow-Blocktown channery loams, 25 to 65 percent slopes	.32	No	C	Well drained

SOURCE AND ACCURACY
Parcel boundary — Jurisdiction gis — Maryland Department of Planning — MD MAP parcel boundaries — Plat...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 60'

0 60 120 180 240 FEET



KEALEE

Design, build, deliver

PROJECT
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
ADDRESS
6751 Columbia Gateway Dr, Columbia, Maryland, 21046
JURISDICTION
Howard County, Maryland
ZONE

SHEET
L-100 — Landscape and Tree Canopy Plan
SCALE
1" = 60'
STATUS
PRELIMINARY
SHEET NO.
5 OF 5
COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

PROFESSIONAL RESPONSIBILITY
Qualified Environmental Professional
certifies: tree_conservation

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope.

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	L-100 — Landscape and Tree Canopy Plan
DISCIPLINE	Landscape Architect / Qualified Professional
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

TAX / PARCEL ID	140648965	
GROSS TRACT AREA	269,740 SF	6.192 AC
LOT FRONTAGE	—	576.41'
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	—	—
SIDE YARD	—	—
REAR YARD	—	—
COVERAGE	MAX	PROPOSED
LOT COVERAGE	—	0.4%
BUILDING FOOTPRINT	—	1,200 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	269,740 SF
2. Area of dwelling	1,200 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	269,740 SF
6. TOTAL AREA DISTURBED	13,257 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH HOWARD COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
6751 Columbia Gateway Dr, Columbia, Maryland, 21046	—	—	—	—	20'	2

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total — 0.7% Impervious	269,740	6.192	
PRE-DEVELOPMENT	SF		C
Pervious — lawn / meadow	269,740		0.25
Composite C			0.250
POST-DEVELOPMENT	SF		C
Proposed roof	1,200		0.95
Driveway and on-lot paving	574		0.95
Pervious — lawn	267,966		0.25
Composite C			0.255
Flow path			1001 ft
Flow path slope			1.4 %
Time of concentration (Kirpich)			8.2 min
Rainfall intensity, 10-yr			5.71 in/hr
Peak discharge PRE, Q = CIA			9.84 cfs
Peak discharge POST, Q = CIA			9.02 cfs
INCREASE TO BE MANAGED			0.18 cfs
Rv = 0.05 + 0.0091			0.059
Water quality volume WQv			1,257 cf

Time of concentration by Kirpich over a 1001 ft flow path at 1.4%, taken from the 2 ft contours across the catchment. A 5-minute floor applies.
Rainfall intensity 5.71 in/hr for the 10-year storm at a 6.2 minute duration, NOAA Atlas 14 Vol. 2 Ver. 3, PFDS point estimate, 39.1842 N 76.8016 W.
Water quality volume on a 1.0 in rainfall, Maryland Stormwater Design Manual, Vol. 1 & 2 — WQv on a 1.0 in rainfall.
Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed areas.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION. FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY TEST PIT BEFORE CONSTRUCTION.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS COMPILED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED. THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION, NOT AN OPEN DITCH SECTION. GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSED CURB AT THE APRON PER HOWARD COUNTY STANDARD. ONLY LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE

DATE

SEAL

LICENSE NO.

EXPIRATION DATE

REVISIONS

— NONE —

SOURCE AND ACCURACY
Parcel boundary — Jurisdiction gis — Maryland Department of Planning — MD IMAP parcel boundaries — Plat...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 60'

0 60 120 180 240 FEET